

Simulation Input Uncertainty Module (SIUM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Governed Space: Simulation Input Uncertainty

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Canonical Language: English (UCL™)

Governed Space

Simulation Input Uncertainty

Canonical Definition

Simulation Input Uncertainty Module (SIUM) defines the governance framework through which uncertainty associated with Simulation Inputs is identified, classified, bounded, quantified or qualitatively characterized where appropriate, propagated into simulation execution, and preserved through downstream interpretation so that uncertainty in the input basis cannot disappear merely because the simulation produces precise outputs.

SIUM establishes the final internal governance layer of SIDS by governing the transition:

Identified + Representative + Transformed + Temporally Governed Inputs → Input
Uncertainty Identification → Uncertainty Characterization → Dependency
Assessment → Uncertainty Propagation Requirements → Simulation Input
Uncertainty Status

Its purpose is not to eliminate uncertainty.

Its purpose is to ensure that uncertainty remains visible, attributable, bounded, and operationally consequential wherever it can materially affect

simulation behavior or conclusions.

Operational Role

SIUM serves as the fifth and final internal governance space of the Simulation Input & Data Standard (SIDS).

The preceding modules establish:

SIIM — What information is entering the simulation?

SDRM — What reality can that information legitimately represent?

SDTM — What happened to that information before simulation use?

STDM — Is that information temporally appropriate for the reality and dynamics being simulated?

SIUM asks:

What uncertainty remains in the complete Simulation Input basis, where does that uncertainty originate, how may uncertainties interact, and what must remain visible when the simulation uses those inputs?

The governed progression is:

Simulation Input Basis → Uncertainty Sources → Uncertainty Classification → Uncertainty Magnitude / Character → Dependency → Materiality → Propagation Requirement → Input Uncertainty Status → Simulation Use

Module Operational Space

SIUM governs: input uncertainty identification, uncertainty source attribution, measurement uncertainty, observational uncertainty, sampling uncertainty, representativeness uncertainty, transformation-induced uncertainty, temporal uncertainty, estimation uncertainty, interpolation uncertainty, extrapolation uncertainty, imputation uncertainty, synthetic-input uncertainty, parameter uncertainty, classification uncertainty, source-dependent uncertainty, missing-information uncertainty, uncertainty range, uncertainty distribution, uncertainty bounds, qualitative uncertainty, quantitative uncertainty, known uncertainty, partially characterized uncertainty, unknown or unresolved uncertainty, uncertainty dependency, correlated uncertainty, compounded uncertainty, uncertainty materiality, uncertainty sensitivity, uncertainty propagation requirements, uncertainty aggregation, uncertainty conditions, uncertainty limitations, uncertainty status, uncertainty reassessment, and uncertainty traceability.

Module Function

SIUM exists to prevent: uncertain inputs being represented as certain, estimated values being treated as observations, measurement uncertainty disappearing during ingestion, transformation-induced uncertainty being ignored, temporal uncertainty being lost, multiple uncertainties being assumed independent without basis, uncertainty being hidden by numerical precision, uncertainty ranges being collapsed into single values without justification,

incomplete information being interpreted as certainty, uncertainty being averaged away, model inputs being presented as exact merely because the simulation requires exact numerical values, simulation outputs appearing more certain than their input basis legitimately supports, and downstream users losing visibility into material uncertainty inherited from Simulation Inputs.

Simulation Input Uncertainty

Simulation Input Uncertainty is the material uncertainty associated with the identity, value, state, origin, representation, transformation, timing, completeness, or applicability of information entering a simulation.

It may concern:

What the value is

Whether the value is correct

Whether the value represents the intended reality

When the value applies

How the value was produced

How complete the information is

or:

How confidently the input may be relied upon within the simulation context.

Uncertainty Is Not Error

SIMULOS® establishes:

Uncertainty does not necessarily mean that an input is wrong.

An input may be the best available representation while still containing material uncertainty.

For example:

Temperature = 20°C ± 1°C

does not mean the measurement is invalid.

It means the information basis contains a bounded uncertainty.

Error vs Uncertainty

SIUM distinguishes:

Known Error

from:

Uncertainty.

A known incorrect value may require correction.

An uncertain value may remain legitimate if the uncertainty is sufficiently understood and governed.

Uncertainty Source

Every material uncertainty should be attributable, where possible, to an identifiable source.

Possible sources include: measurement process, sensor precision, sampling, source reliability, missing information, estimation, transformation, interpolation, extrapolation, temporal misalignment, synthetic generation, model-derived inputs, classification, environmental variability, or unresolved knowledge limitations.

Measurement Uncertainty

Measurement systems have finite accuracy and precision.

A measured value may therefore represent:

Observed Value $\pm$ Measurement Uncertainty

rather than an exact physical state.

Where material, the uncertainty should remain connected to the input.

Observational Uncertainty

Some phenomena cannot be observed perfectly.

Observational uncertainty may arise from: incomplete visibility, ambiguous detection, limited sensor coverage, human observation, environmental interference, or indirect observation.

Sampling Uncertainty

A sample represents only part of a broader population, period, environment, or state space.

Sampling uncertainty asks:

How uncertain is the inference from the observed sample to the reality the simulation intends to represent?

SDRM governs representativeness.

SIUM governs the uncertainty remaining within that representation.

Representativeness Uncertainty

A dataset may be accepted as sufficiently representative while still containing uncertainty about how completely it captures: rare conditions, subpopulations, environmental variation, extreme behavior, or future conditions.

Acceptance under SDRM does not eliminate that uncertainty.

Transformation-Induced Uncertainty

SDTM may introduce uncertainty through: filtering, aggregation, interpolation, extrapolation, imputation, normalization, data fusion, synthetic supplementation, or derived-variable construction.

SIUM consolidates these uncertainty contributions into the governed Simulation Input basis.

Temporal Uncertainty

STDM may identify uncertainty involving: timestamps, synchronization, latency, event timing, temporal alignment, applicability periods, drift, or regime transition.

SIUM ensures material temporal uncertainty remains part of the complete uncertainty basis.

Estimation Uncertainty

Estimated inputs should preserve uncertainty associated with the estimation method.

Conceptually:

Observed Basis + Estimation Method → Estimated Input + Estimation Uncertainty
Interpolation Uncertainty

Values generated between observations may contain uncertainty dependent upon: distance between observations, variability, interpolation method, system dynamics, and missing-data structure.

Extrapolation Uncertainty

Extrapolation extends beyond observed information.

Uncertainty may increase as extrapolation moves further from the observed basis.

SIMULOS® establishes:

Numerical continuation beyond observed data does not establish equivalent informational confidence.

Imputation Uncertainty

An imputed value may be necessary for simulation execution.

But the simulation should preserve that:

Imputed Value $\neq$ Observed Value

and that the imputation carries uncertainty derived from its method and basis.

Synthetic Input Uncertainty

Synthetic information may represent plausible conditions without being direct observation.

Where synthetic inputs are used, SIUM governs uncertainty associated with: generation assumptions, generator limitations, distribution fit, scenario coverage, and divergence from observed reality.

Parameter Uncertainty

Simulation Inputs may include parameters whose exact values are unknown.

A parameter may therefore be represented as:

Fixed Estimate

Range

Distribution

Scenario Set

or:

Unknown within defined bounds.

The representation should reflect the available basis rather than artificial precision.

Classification Uncertainty

Some inputs depend upon classification.

For example:

an object may be classified as:

Pedestrian

with:

92% confidence.

Where classification uncertainty can materially affect simulation behavior, it should remain visible.

Missing-Information Uncertainty

Missing information creates uncertainty even where no substitute value is inserted.

SIMULOS® establishes:

Unknown does not become known merely because the simulation can execute without the missing information.

Known Uncertainty

A Known Uncertainty has an identifiable source and sufficiently characterized effect or range.

Example:

Sensor accuracy $\pm 0.5^{\circ}\text{C}$ Partially Characterized Uncertainty

A Partially Characterized Uncertainty is known to exist but cannot be fully quantified or bounded.

It should remain explicitly represented.

Unresolved Uncertainty

An Unresolved Uncertainty exists where material uncertainty is known but its magnitude, distribution, or consequences cannot yet be sufficiently established.

Unresolved uncertainty may require: restriction, sensitivity analysis, scenario expansion, additional evidence, or downstream limitation.

Unknown Uncertainty

Not every uncertainty can be anticipated.

SIUM does not claim to identify unknowable conditions.

It requires that known limitations in the ability to characterize uncertainty remain visible.

Quantitative Uncertainty

Where appropriate, uncertainty may be expressed through: range, interval, probability distribution, confidence interval, variance, standard deviation, tolerance, error bound, or another justified quantitative representation.

Qualitative Uncertainty

Not all uncertainty should be forced into numerical form.

Qualitative states may include:

Low

Moderate

High

Material but Unquantified

Unresolved

or another governed classification.

No False Quantification Principle

SIMULOS® establishes:

Uncertainty should not be assigned artificial numerical precision where the available information cannot legitimately support it.

Uncertainty Bounds

Where uncertainty can be bounded, the relevant range should be identified.

For example:

Input $X \in [8.5, 10.2]$

A bounded range should reflect the available evidence rather than an arbitrary convenience range.

Unbounded or Open Uncertainty

Some uncertainty may not have a defensible upper or lower bound.

Such uncertainty should not be converted into a narrow interval merely to satisfy a simulation interface.

Point Estimate Governance

A simulation may technically require a single value.

Where a point estimate represents an uncertain quantity, the record should preserve:

Point Estimate + Uncertainty Basis

rather than allowing the estimate to become indistinguishable from exact observation.

No Point-Estimate Certainty Principle

A single numerical input does not become certain merely because the simulation engine accepts only one number.

Uncertainty Dependency

Uncertainties may be dependent.

For example:

multiple inputs derived from the same sensor may share the same measurement uncertainty.

Treating them as independent may underestimate total uncertainty.

Correlated Uncertainty

Where uncertainty in one input is materially correlated with another, the relationship should be preserved where feasible.

Conceptually:

Uncertainty A ↔ Uncertainty B

rather than:

Uncertainty A + Independent Uncertainty B Shared-Source Uncertainty

Multiple inputs may inherit uncertainty from a common source.

For example:

Input A

Input B

and:

Input C

may all depend upon:

Sensor S

A failure or bias in Sensor S may therefore affect all three inputs simultaneously.

No Independence-by-Default Principle

SIMULOS® establishes:

Input uncertainties must not be assumed independent merely because they are represented as separate variables.

Compounded Uncertainty

Multiple uncertainties may interact.

Conceptually:

Measurement Uncertainty + Transformation Uncertainty + Temporal Uncertainty +
Parameter Uncertainty

may create a broader uncertainty condition than any individual contribution alone.

Uncertainty Aggregation

Where multiple uncertainty contributions are aggregated, the aggregation method should be: identifiable, appropriate, reproducible where required, and bounded by known dependencies.

No Simple Addition Rule

Uncertainty components should not automatically be added, averaged, or multiplied without a justified methodological basis.

Uncertainty Materiality

Not every uncertainty requires the same governance intensity.

An uncertainty is material where it could reasonably affect: simulation behavior, scenario outcome, output interpretation, validation, operational decision, or reliance.

Materiality Is Context-Dependent

An uncertainty of:

± 1 mm

may be irrelevant in one simulation and critical in another.

Therefore:

Uncertainty Magnitude $\neq$ Uncertainty Materiality

Materiality depends upon the Simulation Purpose and the sensitivity of the simulated system.

Uncertainty Sensitivity

A simulation may respond differently to uncertainty across inputs.

Some inputs may have little effect.

Others may produce major output changes.

SIUM therefore supports identification of:

Uncertainty-Sensitive Inputs

where uncertainty requires stronger downstream attention.

Sensitivity Requirement

Where material uncertainty exists, downstream simulation design may require sensitivity analysis.

The relevant later SIMULOS® governance spaces determine how that analysis is executed.

SIUM establishes that the uncertainty requiring examination must not disappear before reaching them.

Uncertainty Propagation

Input uncertainty may propagate through:

Simulation Inputs

↓

Simulation Model

↓

Simulation Execution

↓

Simulation Outputs

↓

Simulation Interpretation

↓

Operational Reliance

SIUM governs the input-side uncertainty basis and establishes propagation requirements for later SIMULOS® stages.

Uncertainty Does Not End at Input Boundary

SIMULOS® establishes:

Material uncertainty attached to an input remains relevant until a governed downstream process demonstrates why it no longer materially affects the

simulation result.

No Uncertainty Erasure Principle

A transformation, model, or simulation execution must not silently erase material uncertainty merely because the downstream output is deterministic.

Deterministic Model $\neq$ Certain Result

A deterministic simulation may produce exactly the same output from exactly the same input.

That does not mean the simulated result is certain where the input itself is uncertain.

Precision vs Certainty

A simulation may produce:

0.87346192

from uncertain input data.

The number of decimal places does not determine the certainty of the result.

No Precision-to-Certainty Principle

SIMULOS® establishes:

Computational precision must not be represented as evidentiary certainty.

Uncertainty Scenario Representation

Where uncertainty cannot be adequately represented through a single input, the simulation may require multiple input conditions.

For example:

Low Estimate

Central Estimate

High Estimate

or:

Multiple Probability-Based Samples

SIUM establishes the uncertainty basis requiring such treatment.

Boundary Conditions

Uncertain inputs may require simulation at boundary values.

This can reveal whether conclusions remain stable across plausible input

ranges.

Worst-Credible Input Condition

Where consequence is high, the simulation may need to examine credible adverse input combinations.

SIUM identifies the uncertainty basis.

Scenario governance determines how those conditions are represented.

Uncertainty Interaction With Scenario Design

SIUM connects directly to the subsequent Simulation Scenario Standard (SSS).

Uncertainty may require scenarios that explore: plausible ranges, adverse combinations, edge conditions, uncertainty distributions, alternative input states, and unresolved input assumptions.

SIUM does not design those scenarios.

It establishes what uncertainty the scenario design must be capable of addressing.

Uncertainty Register

Material input uncertainties should be capable of being preserved within a Simulation Input Uncertainty Register.

The register may identify: uncertainty identifier, affected input, uncertainty source, uncertainty type, magnitude or characterization, dependency, materiality, downstream relevance, condition, and current status.

Uncertainty Identifier

A material uncertainty may receive a persistent identifier.

For example:

SIU-0047

This allows later simulation records to reference the same uncertainty across: scenario design, execution, output interpretation, validation, and reliance.

Uncertainty Status

Possible uncertainty statuses may include:

Characterized

Partially Characterized

Bounded

Conditionally Bounded

Unresolved

Material Unquantified

or:

Not Material for Defined Simulation Use Uncertainty Acceptance

An uncertainty may be accepted for simulation use where: it is sufficiently understood, its effect is bounded, required downstream treatment exists, and the Simulation Purpose permits reliance upon that uncertainty level.

Acceptance does not mean the uncertainty disappears.

Conditional Uncertainty Acceptance

A Simulation Input may remain usable subject to conditions.

For example:

Input accepted only if sensitivity testing covers $\pm 10\%$ variation.

Uncertainty Restriction

An input may be restricted where uncertainty prevents broader use.

For example:

Suitable for exploratory simulation only.

but:

Not suitable for safety-critical operational reliance.

Uncertainty Rejection

Where uncertainty is too great or too poorly understood for the intended simulation purpose, the input may be rejected.

Uncertainty Undetermined

Where the available information is insufficient to assess uncertainty, the correct state may be:

Uncertainty Undetermined

rather than an assumed acceptable level.

Uncertainty Threshold

A simulation may define thresholds beyond which input uncertainty requires: additional evidence, broader scenarios, stronger sensitivity testing, restriction, or rejection.

Thresholds should derive from the Simulation Purpose and consequence rather than convenience.

Uncertainty Budget

Where appropriate, a simulation may establish an Input Uncertainty Budget defining how much uncertainty can be tolerated across material input categories.

The budget should not permit one uncertainty to disappear merely because another input is highly precise.

No Precision Compensation Principle

SIMULOS® establishes:

High certainty in one Simulation Input cannot automatically compensate for material uncertainty in another input upon which the simulated outcome critically depends.

Uncertainty Concentration

A simulation may contain many high-quality inputs but depend critically upon one highly uncertain variable.

SIUM therefore assesses uncertainty according to influence, not merely average input quality.

Weakest Critical Input Principle

Where an outcome materially depends upon a highly uncertain critical input, the simulation conclusion should remain bounded by that uncertainty.

Uncertainty Change

Input uncertainty may change when: new evidence becomes available, sensors are recalibrated, source reliability changes, transformation logic changes, temporal distance increases, missing values increase, new environments are introduced, or Simulation Scope changes.

Uncertainty Reassessment Trigger

Material changes should trigger reassessment of the affected uncertainty basis.

Scope Expansion and Uncertainty

Expanding Simulation Scope may expose uncertainty that was immaterial under the original scope.

For example:

an input sufficiently certain for:

normal operating conditions

may be insufficiently characterized for:

extreme operating conditions.

Version Change and Uncertainty

A new input version should not automatically inherit the uncertainty characterization of its predecessor.

The new version may require reassessment where: source, measurement, transformation, temporal basis, or distribution

has materially changed.

Uncertainty Traceability

SIUM establishes:

Simulation Input

↓

Uncertainty Source

↓

Uncertainty Type

↓

Magnitude / Characterization

↓

Dependency

↓

Materiality

↓

Propagation Requirement

↓

Condition / Restriction

↓

Uncertainty Status

↓

Simulation Use Input Uncertainty Basis

The final SIUM result should establish a coherent Simulation Input Uncertainty Basis for the complete Simulation Input Set.

This basis should identify: material uncertainties, their origins, their relationships, their characterization, unresolved uncertainty, downstream treatment requirements, and limitations upon simulation use.

Input-Set Uncertainty

Individual inputs may each have acceptable uncertainty while their combination creates material uncertainty.

SIUM therefore distinguishes:

Input-Level Uncertainty

from:

Input-Set Uncertainty.

Input-Set Uncertainty Coherence

The complete uncertainty basis should account for: shared sources, dependencies, correlations, transformation chains, temporal relationships, and interacting uncertainty.

Simulation Input Readiness

After SIUM, the complete SIDS input basis may receive an overall readiness status such as:

Simulation Input Ready

Conditionally Simulation Input Ready

Partially Simulation Input Ready

Simulation Input Not Ready

or:

Simulation Input Readiness Undetermined

This status represents completion of the SIDS input-governance sequence.

Input Readiness Does Not Mean Simulation Validity

SIMULOS® establishes:

Simulation Input Readiness confirms only that the input basis is sufficiently governed for its defined simulation use. It does not establish that the model, scenarios, execution, outputs, or simulation itself are valid.

Those questions belong to subsequent SIMULOS® Parent Standards.

Minimum Implementation Framework

1. Identify Material Input Uncertainties

Determine which Simulation Inputs contain uncertainty capable of materially affecting simulation behavior or interpretation. 2. Identify Uncertainty Sources

Bind each material uncertainty to its known or suspected origin. 3. Classify Uncertainty

Determine whether uncertainty is associated with: measurement, observation, sampling, representativeness, transformation, time, estimation, interpolation, extrapolation, imputation, synthetic generation, parameterization, classification, or another relevant source.

4. Characterize Uncertainty

Use justified: quantitative bounds, distributions, qualitative classifications, or explicit unresolved status.

5. Assess Dependencies

Determine whether material uncertainties are: independent, correlated, shared-source, sequentially dependent, or otherwise linked.

6. Assess Materiality

Determine whether each uncertainty can materially affect the Simulation Purpose, behavior, outputs, or downstream reliance. 7. Establish Propagation Requirements

Identify which uncertainties must remain visible through subsequent simulation stages. 8. Establish Conditions and Restrictions

Define limitations upon simulation use where uncertainty remains material. 9. Assign Uncertainty Status

Classify material uncertainties as: Characterized, Partially Characterized, Bounded, Conditionally Bounded, Unresolved, Material Unquantified, or Not

10. Assess Input-Set Uncertainty

Evaluate interactions across the complete Simulation Input basis. 11.
Determine Simulation Input Readiness

Establish whether the governed input basis is: Ready, Conditionally Ready,
Partially Ready, Not Ready, or Readiness Undetermined.

12. Preserve Uncertainty Traceability

Maintain:

Input → Uncertainty → Source → Characterization → Dependency → Materiality →
Propagation → Condition → Status → Simulation Use

Governance Outputs

SIUM may produce: Simulation Input Uncertainty Register, Input Uncertainty Identifier, Uncertainty Source Record, Measurement Uncertainty Record, Observational Uncertainty Record, Sampling Uncertainty Record, Representativeness Uncertainty Record, Transformation-Induced Uncertainty Record, Temporal Uncertainty Record, Estimation Uncertainty Record, Interpolation Uncertainty Record, Extrapolation Uncertainty Record, Imputation Uncertainty Record, Synthetic Input Uncertainty Record, Parameter Uncertainty Record, Classification Uncertainty Record, Missing-Information Uncertainty Record, Quantitative Uncertainty Assessment, Qualitative Uncertainty Assessment, Uncertainty Bound, Uncertainty Distribution Record, Unresolved Uncertainty Record, Uncertainty Dependency Map, Correlated Uncertainty Record, Shared-Source Uncertainty Record, Compounded Uncertainty Assessment, Uncertainty Materiality Assessment, Uncertainty Sensitivity Signal, Uncertainty Propagation Requirement, Input Uncertainty Budget, Uncertainty Condition Record, Uncertainty Restriction Record, Uncertainty Status, Input-Set Uncertainty Assessment, Simulation Input Uncertainty Basis, Simulation Input Readiness Status, Uncertainty Reassessment Trigger, and Simulation Input Uncertainty Traceability Map.

Use Case 1 — Autonomous Robot Simulation

Scenario

A robot-navigation simulation receives: object-position measurements with sensor error, estimated surface-friction values, partially observed pedestrian movement, and interpolated localization data.

Each input is technically usable.

However, each contains a different uncertainty basis. Application

SIUM identifies: measurement uncertainty in object position, estimation uncertainty in surface friction, observational uncertainty in pedestrian movement, and interpolation uncertainty in localization.

Dependencies between localization and object-position measurements are also identified because both depend partly upon the same sensor system. Result

The input basis becomes:

Conditionally Simulation Input Ready

with a requirement that material uncertainty be propagated into subsequent scenario and execution governance rather than treating the nominal input values as exact reality. Use Case 2 — Energy Infrastructure Simulation

Scenario

A long-term energy simulation uses: historical demand, future population projections, weather forecasts, estimated technology adoption, and assumed future industrial demand.

The simulation produces precise annual energy-demand figures. Application

SIUM distinguishes:

Observed Historical Inputs

from:

Projected Inputs

Estimated Inputs

and:

Assumed Inputs.

The largest material uncertainty is found in future industrial demand.

Sensitivity to this input is high. Result

The simulation may continue, but the uncertainty basis requires downstream scenarios covering materially different industrial-demand conditions.

The final simulation cannot legitimately present the central numerical trajectory as a certain future state.

Architectural Position

Simulation Input Uncertainty is the fifth and final internal governance space of the Simulation Input & Data Standard (SIDS).

The complete SIDS progression is:

Simulation Input Identification

↓

Simulation Data Representativeness

↓

Simulation Data Transformation

↓

Simulation Temporal Data

↓

Simulation Input Uncertainty

Together, the five modules answer:

What information is entering the simulation?

↓

What reality can it legitimately represent?

↓

What happened to it before simulation use?

↓

Is it temporally appropriate?

↓

What uncertainty remains in the complete input basis?

The resulting:

Simulation Input Uncertainty Basis

and:

Simulation Input Readiness Status

provide the governed input foundation required by subsequent SIMULOS® lifecycle spaces.

The next Parent Standard, Simulation Scenario Standard (SSS), governs which scenarios must be designed to examine expected conditions, variations, edge cases, stresses, uncertainty, and materially relevant alternatives.

Foundational Principle

Simulation Inputs become uncertainty-governed only when material uncertainty remains identifiable from its source through its characterization, dependencies, conditions, and downstream propagation requirements, so that computational precision cannot transform uncertain information into an appearance of certain reality.

Canonical Closing Statement

Simulation Input Uncertainty Module completes the Simulation Input & Data Standard by governing the uncertainty that remains after Simulation Inputs have been identified, assessed for representativeness, transformed, and temporally governed. Through uncertainty-source attribution, classification, quantitative and qualitative characterization, dependency and correlation assessment, materiality, sensitivity, propagation requirements, conditions, restrictions, reassessment, Input-Set Uncertainty, readiness determination, and complete uncertainty traceability, SIUM ensures that uncertainty entering a simulation cannot disappear merely because the simulation can process the data or produce precise numerical outputs. The module establishes the final governed Simulation Input Uncertainty Basis from which subsequent SIMULOS® scenario design and simulation execution can legitimately proceed.

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